Harmonic Analysis for Edge Based Smart Grids: Feature Engineering and Network-Efficient Power Estimation

Leonardo Santiago Melgaço Silva[1], **Gabriel Arquelau Pimenta Rodrigues**[1], **Mariana Cabral Falqueiro**[2], **Geraldo Pereira Rocha Filho**[3], **André Luiz Marques Serrano**[1], **Vinícius Pereira Gonçalves**[1]

[1]Electrical Engineering Department (ENE) – University of Brasilia, Brasilia, DF, Brazil

[2]IFB – Instituto Federal de Brasilia – Distrito Federal, DF – Brazil

[3]Department of Exact and Technological Sciences (DCET) – State University of Southwest Bahia, Vitória da Conquista, BA, Brazil

252110506@aluno.unb.br, gabriel.arquelau@redes.unb.br,mariana.cabral@unb.br, geraldo.rocha@uesb.edu.br,andrelms@unb.br,vpgvinicius@unb.br

Abstract. *High frequency electrical measurements enable harmonic analysis of load behavior in smart grid environments, supporting advanced applications such as non intrusive load monitoring (NILM). While prior work has focused on complex machine learning models, their computational cost limits deployment in edge computing scenarios.*

In this paper, we investigate the role of feature engineering in harmonic based power estimation, with emphasis on network efficiency and edge intelligence. We adopt a leakage aware experimental protocol and evaluate the contribution of harmonic features to predictive performance.

Our results show that the fundamental harmonic (h1) alone explains nearly all the variance in active power, achieving performance equivalent to models using the full harmonic spectrum. This indicates that the underlying mapping is inherently low complexity and can be effectively captured using lightweight models.

These findings demonstrate that harmonic analysis combined with feature centric design enables accurate and network efficient power estimation, significantly reducing computational cost, latency, and data transmission requirements. This makes the proposed approach particularly suitable for scalable, edge based smart grid systems.

Keywords: Harmonic Analysis; Edge Computing; Smart Grids; Feature Engineering; Network Efficiency; Non-Intrusive Load Monitoring.

1. Introduction

Smart grids increasingly rely on data driven intelligence to monitor, analyze, and optimize electrical energy consumption. In this context, high frequency measurements enable detailed harmonic analysis of electrical signals, supporting applications such as non intrusive load monitoring (NILM) and real time energy analytics.

Despite recent advances, most existing approaches focus on improving predictive performance through increasingly complex machine learning and deep learning models.

While effective, these approaches often impose significant computational and communication overhead, limiting their applicability in edge computing environments where resources are constrained and low latency processing is required.

This paper investigates an alternative perspective based on feature engineering, focusing on the role of harmonic features in power estimation. In particular, we analyze how the fundamental harmonic component can capture most of the information required to model active power, enabling highly efficient and low complexity solutions.

We adopt a leakage aware experimental protocol using high frequency electrical measurements and evaluate the contribution of harmonic features to predictive performance. Our results show that the fundamental harmonic (h1) alone explains nearly all the variance in active power, achieving performance equivalent to models using the full harmonic spectrum.

These findings suggest that the learning task is inherently low complexity and that feature centric design can replace model centric approaches in this context. From a systems perspective, this enables network efficient architectures, where data transmission is minimized and inference is performed locally at the edge, reducing latency and improving scalability.

1.1. Contributions

The main contributions of this work are fourfold. First, we provide a feature centric analysis of harmonic representations specifically tailored for power estimation in smart grids. To ensure a reliable evaluation of these representations, we implement a leakage aware experimental protocol that prevents data contamination. Furthermore, we demonstrate that the fundamental harmonic (h_1) is the dominant feature in the prediction task, effectively capturing the core electrical behavior. Finally, we analyze the edge and network implications of our findings, offering insights into the design of more efficient and scalable smart grid systems.

2. Related Work

Recent works in energy analytics and non intrusive load monitoring (NILM) have widely adopted machine learning and deep learning techniques to model complex electrical patterns. These approaches typically focus on improving predictive accuracy through increasingly sophisticated models, such as convolutional neural networks and sequence to sequence architectures [Zhang et al. 2024][Regan et al. 2022]. However, this pursuit of accuracy often comes at the cost of higher computational complexity, requiring significant processing power and memory [Goncalves et al. 2025][Yue and et al. 2023].

In parallel, the emergence of edge computing has motivated research on lightweight AI models capable of operating under resource constraints, with reduced latency and bandwidth usage [Zonzini et al. 2025]. In this distributed context, model efficiency becomes as important as predictive performance. Several studies have explored model compression and pruning techniques to deploy intelligence directly at the smart meter level [Magnani et al. 2024].

Despite these efforts, limited attention has been given to the role of feature dominance in simplifying learning tasks. While some works explore harmonic signatures

[de Oliveira Barbosa et al. 2017], they often use them as inputs for heavy classifiers. In contrast to prior work that emphasizes model complexity, this paper focuses on understanding how a single highly informative feature the fundamental harmonic can explain most of the variance in active power. This shifts the design perspective from model-centric to feature centric optimization, particularly relevant for networked edge systems. Ultimately, this work departs from prior approaches by demonstrating that feature dominance can eliminate the need for complex models altogether.

3. System Model and Smart Metering Perspective

We consider a three tier architecture composed of smart meters, edge gateways, and a cloud backend.

High-frequency signals are captured at the smart meter and transformed into harmonic features. Edge devices perform local inference, transmitting only compact outputs to the cloud, reducing bandwidth usage and latency.

As illustrated in Figure 1, the proposed architecture shifts computation from the cloud to the edge, reducing communication overhead and enabling real time inference. By emphasizing local feature extraction and lightweight inference, the proposed system model aligns with the constraints of large scale smart grid deployments, where bandwidth, computational resources, and energy consumption are critical factors.

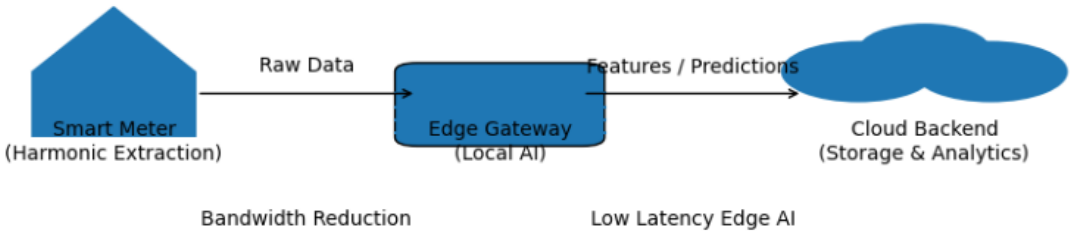

Figure 1. Proposed edge oriented smart grid architecture. High frequency electrical signals are acquired at the smart meter, where harmonic features are extracted. Raw data is locally processed and forwarded to an edge gateway, where lightweight machine learning models perform real time inference. Only processed features or prediction outputs are transmitted to the cloud backend, reducing bandwidth usage, enabling low latency operation, and improving privacy.

4. Materials and Methods

This work uses a high frequency NILM dataset that contains harmonic level electrical features [Dinar et al. 2025]. The dataset was acquired using a measurement system based on the ATM90E36A energy metering integrated circuit, which simultaneously monitors aggregate household consumption and individual appliance consumption. The measured appliances are a hair dryer, an electric water heater, a hair straightener, a fridge, an iron, a screen, a laptop charger, and a lamp. The dataset is divided into measurement sessions, collected across multiple recording periods. Each sensing node records RMS voltage and current, active and apparent power, power factor, and the magnitudes of current harmonics from the 1st to the 32nd order, computed using the IC's onboard Discrete Fourier

Transform engine applied to an eight kHz sampled signal. Measurements are logged at two second intervals.

Figure 2 illustrates the complete experimental pipeline adopted in this study, from high frequency data acquisition to model evaluation.

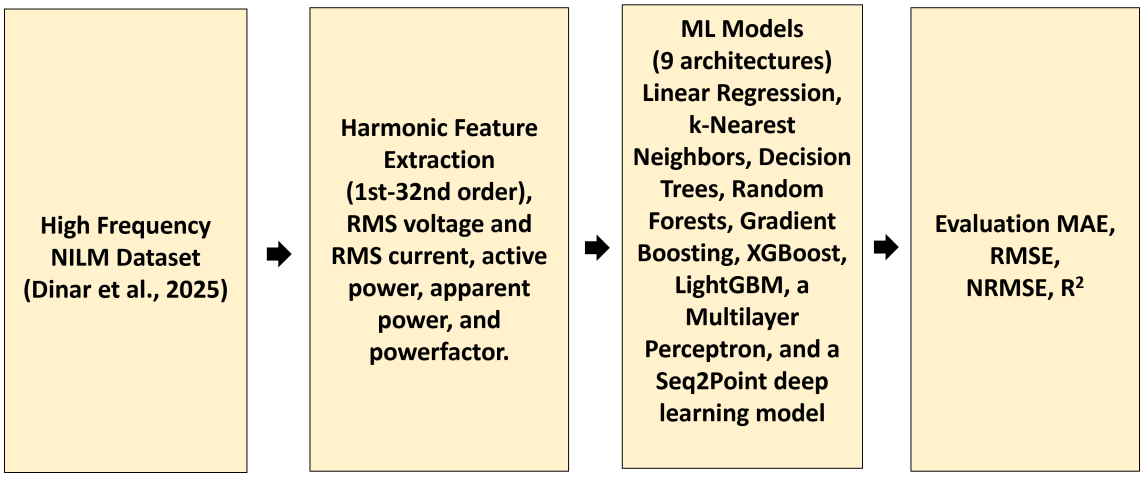

Figure 2. Experimental pipeline adopted in this study, including dataset acquisition, feature normalization, model training, and performance evaluation.

4.1. Dataset and Feature Space

The dataset used in this study is a high frequency NILM dataset based on harmonic feature analysis, originally introduced by [Dinar et al. 2025]. The experiments are conducted using a high frequency electrical dataset composed of synchronized measurements of electrical quantities and harmonic components. For each acquisition session and sensor phase channel, the dataset provides root mean square current (I_{rms}), root mean square voltage (V_{rms}), power factor, apparent power, and the magnitudes of the first 32 current harmonics. Appliance- level ground truth is not available in the dataset; therefore, the experimental analysis focuses on the contribution of harmonic features to electrical power modeling rather than supervised load disaggregation.

All measurements are acquired using a sensing system based on the ATM90E36A energy metering integrated circuit, which computes harmonic components through an onboard Discrete Fourier Transform applied to an 8 kHz sampled signal. Feature values are logged at two second intervals and collected across multiple measurement sessions and phases, enabling the analysis of both steady state and dynamic operating conditions.

The dataset is partitioned at the session level, ensuring that samples from the same acquisition session do not appear simultaneously in training and testing sets. Feature normalization is performed using statistics computed exclusively from the training data and then applied to the test data.

5. Experimental Setup

All experiments are conducted using Python version 3.12.12 and executed on Google Colab. Classical models are implemented using the *scikit-learn* library 1.6.1 version, while deep learning models are developed using PyTorch 2.9.0 version.

Hyperparameter tuning is performed using grid search or Optuna 4.6.0 version, and all experiments use fixed random seeds to ensure reproducibility.

All reported results correspond to deterministic training procedures or fixed random seeds, ensuring full reproducibility of the benchmark and consistent comparisons across models.

5.1. Feature Engineering

We extract harmonic components (h1–h32) from high frequency measurements. To avoid leakage, features directly derived from electrical power equations (e.g., voltage, current, apparent power) are excluded.

5.2. Experimental Protocol

Session based train/test split, Normalization using training data only and evaluation using MAE, RMSE, and R^2.

6. Limitations

The experimental evaluation is limited to a single high frequency dataset and does not assess cross dataset generalization. This choice prioritizes experimental feasibility and reproducibility, reflecting realistic deployment constraints rather than aiming to maximize deep learning performance at any cost. Additionally, due to memory and computational constraints, the Seq2Point model is trained using reduced temporal windows. Although this configuration preserves the core architectural principles, it may limit the model's ability to capture long term temporal dependencies.

7. Results and Discussion

7.1. Quantitative Results

The results confirm that adding more harmonic features does not significantly improve predictive performance compared to using only the fundamental harmonic. This indicates that the learning task is dominated by a single feature, resulting in an inherently low complexity mapping.

Table 1. Performance comparison of feature sets

Feature Set	MAE	RMSE	R^2
h1 only	1.03	1.94	0.9999
All harmonics (h1–h32)	1.03	1.94	0.9999

7.2. Feature Dominance and Linear Mapping

A deeper correlation analysis reveals that the fundamental harmonic (h1) maintains a correlation of approximately 0.99 with the target active power, whereas subsequent harmonics exhibit significantly lower individual contributions. To validate the hypothesis of a low complexity mapping, a Simple Linear Regression model was trained using only h1 as an input.

This lightweight approach achieved a performance equivalent to the most sophisticated models in our benchmark (Table 1), effectively explaining nearly all the variance in

the dataset ($R^2 \approx 0.9999$). These findings provide empirical evidence that, for the task of power estimation in this specific smart grid context, feature dominance allows for the replacement of model centric complexity with feature centric efficiency. Consequently, the use of high parameter models becomes redundant, supporting the deployment of minimal resource algorithms at the network edge.

7.3. Model Complexity Implications

Since h1 alone explains most of the variance, complex models do not provide meaningful performance gains. Lightweight models achieve equivalent results with significantly lower computational cost.

From a physical perspective, this behavior is expected, as the fundamental harmonic captures most of the signal energy, which is directly related to active power.

8. Network Implications for Edge-Based Smart Grid Systems

The results obtained in this study have direct implications for the design and operation of networked smart grid systems, particularly in edge oriented architectures where communication efficiency, scalability, and latency are critical constraints.

First, the use of harmonic aware lightweight models significantly reduces the need for transmitting raw high frequency electrical signals to centralized cloud infrastructures. High frequency measurements, especially when sampled in the kHz range, generate large volumes of data that can quickly saturate network links in large scale deployments. By performing local feature extraction and inference at the edge, only aggregated information or prediction outputs need to be transmitted, leading to substantial reductions in bandwidth consumption.

Second, the reduced computational complexity of lightweight models directly impacts inference latency and enables real time processing at edge devices such as smart meters or local gateways. This low latency capability is essential for time sensitive applications, including demand response, anomaly detection, and real time energy feedback systems. In contrast, deep learning models with higher computational requirements may introduce delays or require offloading to cloud services, increasing communication overhead and system response time.

Third, minimizing data transmission also enhances privacy and security in smart grid environments. Since raw electrical signals can reveal detailed information about user behavior and appliance usage patterns, local processing reduces the exposure of sensitive data over communication networks. This aligns with emerging privacy preserving paradigms in distributed intelligent systems.

From a scalability perspective, the combination of harmonic feature extraction and lightweight inference models enables efficient deployment across large numbers of distributed nodes. As the number of smart meters increases, centralized processing becomes a bottleneck, both in terms of network traffic and computational load. Edge based inference mitigates this issue by distributing computation closer to the data source.

Finally, the findings suggest that feature engineering plays a critical role in network efficient AI systems. By leveraging informative harmonic representations, it is pos-

sible to reduce model complexity without sacrificing predictive performance, leading to more efficient trade offs between accuracy, latency, and communication cost.

Overall, this work demonstrates that harmonic aware machine learning is not only a modeling choice but also a network design strategy, enabling scalable, low latency, and privacy aware intelligent services in modern smart grid infrastructures.

9. Reproducibility

To ensure full reproducibility of the experimental results, all source code used in this study is publicly available. The repository includes data preprocessing routines, model training scripts, and performance evaluation pipelines for all nine evaluated models. A fully executable Google Colab notebook is provided to facilitate replication of the experiments under identical conditions.The complete codebase can be accessed at https://github.com/leonardolsms/Harmonic-Analysis-for-Edge-Based-Smart-Grids.git.

The dataset used in this study is publicly available and was obtained from the high frequency NILM repository maintained by [Dinar et al. 2025] The dataset can be accessed at https://github.com/fariddinar/nilm-dataset (accessed on 20 December 2025).

10. Conclusion

This paper investigated the role of harmonic features in edge based power estimation for smart grid systems. Through a comprehensive experimental analysis, we demonstrated that the fundamental harmonic (h_1) alone explains nearly all the variance in active power. This high correlation allowed a simple Linear Regression model to achieve performance levels equivalent to complex deep learning architectures, such as Seq2Point, with an R^2 of approximately 0.9999.

These results challenge the prevailing assumption that high frequency energy analytics inherently require computationally expensive models. Instead, our findings highlight that strategic feature selection specifically targeting the dominant harmonic is a more effective path toward system efficiency.

From a practical networking perspective, transitioning from a model centric to a feature centric design provides three pivotal advantages for smart grid deployments. First, it enhances bandwidth efficiency, as transmitting only the extracted h_1 component or local predictions significantly reduces data volume compared to raw high frequency signals. Second, the minimal computational overhead of linear models ensures low latency by enabling near instantaneous local inference at the edge a critical requirement for real time demand response and anomaly detection. Finally, this approach promotes scalability and privacy; by processing data locally at the smart meter level, the system reduces the burden on centralized cloud servers and minimizes the exposure of sensitive, high resolution consumption patterns.

In summary, this work demonstrates that harmonic aware machine learning is not just a modeling preference, but a robust network design strategy for scalable and privacy aware intelligent services in modern smart grid infrastructures. Future work will extend this analysis to multi load scenarios and investigate the robustness of these lightweight models under more diverse operating conditions.

11. References

de Oliveira Barbosa, E. H., Oliveira, V. M., Branco, H. M. G. C., and Lemos, M. V. (2017). AplicaÇÃo de redes neurais artificiais para o reconhecimento de assinaturas harmÔnicas de equipamentos eletromÉdicos.

Dinar, F., Paris, S., and Busvelle, E. (2025). Capturing high-frequency harmonic signatures for nilm: Building a dataset for load disaggregation. *Sensors*, 25(15):4601.

Goncalves, E., Moura, P. N., Costa, D., and Alvim, A. (2025). Inteligência artificial verde e quantização em modelos de aprendizagem profunda.

Magnani, S., Tirupathi, S., Doriguzzi-Corin, R., Nedoshivina, L., Braghin, S., and Siracusa, D. (2024). Online learning and model pruning against concept drifts in edge devices. In *2024 IEEE 10th International Conference on Network Softwarization (NetSoft)*, pages 351–356. IEEE.

Regan, J., Saffari, M., and Khodayar, M. (2022). Deep attention and generative neural networks for nonintrusive load monitoring. *The Electricity Journal*, 35(5):107127. Behind the meter strategies for enhancing the electricity grid resilience, reliability, economics, sustainability, and security.

Yue, Z. and et al. (2023). Bert4nilm: A bidirectional transformer model for non-intrusive load monitoring. *IEEE Transactions on Smart Grid*, 14(2).

Zhang, D., Li, X., Li, J., Li, G., Zheng, N., and Wang, Z. (2024). Non-invasive load monitoring based on cnn-lstm-attention. In *2024 IEEE 3rd International Conference on Electrical Engineering, Big Data and Computer (EEBDC)*, pages 568–572. IEEE.

Zonzini, F., Carbone, A., Girolami, A., Testoni, N., De Marchi, L., and Malatesta, M. V. (2025). Lightweight signal processing and edge ai for real-time anomaly detection in iot sensor networks. *Sensors*, 25(21):6629.